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## Dynamic Functional Connectivity in Pediatric Mild Traumatic Brain Injury

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### Abstract

Resting-state fMRI can be used to identify recurrent oscillatory patterns of functional connectivity within the human brain, also known as dynamic brain states. Alterations in dynamic brain states are highly likely to occur following pediatric mild traumatic brain injury (pmTBI) due to the active developmental changes. The current study used resting-state fMRI to investigate dynamic brain states in 200 patients with pmTBI (ages 8-18 years, median = 14 years) at the subacute (~1-week post-injury) and early chronic (~4 months post-injury) stages, and in 179 age- and sex-matched healthy controls (HC). A k-means clustering analysis was applied to the dominant time-varying phase coherence patterns to obtain dynamic brain states. In addition, correlations between brain signals were computed as measures of static functional connectivity. Dynamic connectivity analyses showed that patients with pmTBI spend less time in a frontotemporal

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Declaration of Competing Interest

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Supplementary materials

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default mode/limbic brain state, with no evidence of change as a function of recovery post-injury. Consistent with models showing traumatic strain convergence in deep grey matter and midline regions, static interhemispheric connectivity was affected between the left and right precuneus and thalamus, and between the right supplementary motor area and contralateral cerebellum. Changes in static or dynamic connectivity were not related to symptom burden or injury severity measures, such as loss of consciousness and post-traumatic amnesia. In aggregate, our study shows that brain dynamics are altered up to 4 months after pmTBI, in brain areas that are known to be vulnerable to TBI. Future longitudinal studies are warranted to examine the significance of our findings in terms of long-term neurodevelopment.

## Keywords

cognition; concussion; BOLD; fMRI; dFC; LEiDA; Leading Eigenvector Dynamics

## 1. Introduction

Pediatric mild traumatic brain injury (pmTBI; hereafter used synonymously with concussion) is a commonly occurring neurological condition that may affect brain connectivity (Rausa et al., 2020). A widely used neuroimaging technique to study brain connectivity is resting-state fMRI (rs-fMRI), which measures intrinsic fluctuations in the blood-oxygen level dependent (BOLD) signal throughout the brain during wakeful rest (Snyder & Raichle, 2012). Functional connectivity (FC) describes the level of synchrony between BOLD fluctuations in different regions of the brain (Biswal et al., 1995). In large part due to rs-fMRI research, it is now well established that the brain is organized into intrinsic networks consisting of segregated brain regions that are functionally connected (Raichle, 2015). Considering the ability of rs-fMRI to capture FC on a local as well as global scale, this imaging technique seems well suited for obtaining a systematic understanding of alterations in brain connectivity following pmTBI.

However, previous rs-fMRI research on pmTBI has yielded mixed results. A longitudinal multimodal neuroimaging study has shown increased FC in various intrinsic brain networks in adolescents with less extensive axonal damage (as quantified by diffusion tensor imaging) three months after a sports-related concussion as compared to 24-72 hours post-injury, as well as relative to healthy controls (Manning et al., 2017). Another longitudinal study found increased posterior (precuneus and cerebellum) and decreased anterior (dorso- and ventrolateral prefrontal cortex) FC in adolescent football players within the first seven days after head injury, with no effects at three week follow-up (Murdaugh et al., 2018). A recent cross-sectional study demonstrated increased FC within parts of the cognitive control network (CCN; often found dissociated into different subnetworks with varying nomenclature; Cole & Schneider, 2007; Seeley et al., 2007; Yeo et al., 2011), and both increased and decreased FC between this network and the default mode network (DMN) at one month post-injury, without changes within the DMN itself (Healey et al., 2022). Midline or deep gray matter networks, including the DMN, may be more vulnerable to mTBI due to straining and shearing of white matter connections within these networks (McAllister, 2011; McAllister et al., 2012). Dysregulated DMN activity, and alterations in FC between

DMN and CCN have been proposed as a hallmark of mTBI-related changes in neuronal functioning and are thought to lead to cognitive symptoms through disbalances between internally (mediated by DMN) versus externally (mediated by CCN) focused attentional processes (Healey et al., 2022; Mayer et al., 2011; van der Horn et al., 2019). Other studies reported increased and decreased connectivity within the DMN at approximately one-month post-injury as compared to healthy controls (Borich et al., 2015; Newsome et al., 2016). The relationship between resting-state FC and clinical recovery after pmTBI is as yet unclear (Manning et al., 2017; Murdaugh et al., 2018; Newsome et al., 2016; Shapiro et al., 2021).

The heterogeneity among aforementioned published findings might be explained by, among other things, small sample sizes, the use of cross-sectional instead of longitudinal designs, and/or the use of different computational methods to analyze FC. More importantly, studies so far have computed integral measures of FC for the entire scan duration (e.g., correlation between entire time courses of regions), also known as static FC measures. It is widely accepted that the brain is not static, but dynamic in nature (Calhoun et al., 2014), and therefore study of the dynamic aspect of FC in pmTBI might reveal new insights into its pathophysiology. To our knowledge, no study so far has examined dynamic FC (dFC) after pmTBI.

Following the computation of FC at multiple time points throughout the time course of BOLD fluctuations in different brain regions (i.e., dFC), using, for example, correlation or phase coherence measures, clustering methods can be used to detect time-varying, and recurrent, spatial FC patterns that are known as *dynamic brain states* (Cabral, Kringelbach, et al., 2017; Calhoun et al., 2014). To date, there is a large body of evidence showing that changes in summary measures of these dynamic brain states (e.g., how much time the brain is in a certain state or how often it switches between states) are associated with a wide range of brain diseases (Cohen, 2018). Alterations in dynamic brain states have also been reported in adult mTBI, which were related to clinical outcome measures (van der Horn et al., 2020; Vergara et al., 2018). However, it remains a question how dynamic brain states are affected in pediatric cases and how this relates to clinical outcome. Examining how pmTBI perturbs certain brain states, and how these states are spatially configured in terms of network FC (e.g., within and between DMN and CCN), may provide a step forward in the understanding of pmTBI and its clinical consequences.

The current longitudinal rs-fMRI study therefore set out to examine dFC in a large sample of pmTBI (N=200) and healthy controls (HC; N=179). We hypothesized that the dynamics of brain states may be altered in the sub-acute phase (~ 1 week) post-pmTBI, and recovered in the early chronic phase (~ 4 months). Specifically, we predicted that patients with pmTBI would spend more time in states with high FC between areas within the DMN, and/or high FC between DMN and CCN. In addition, we conducted analyses of static FC, and predicted that the effects of pmTBI would be greatest along the midline and in deep grey matter structures.

## 2. Methods

### 2.1. Participants

Between July 2016 and April 2020, 208 patients with pmTBI (aged 8-18 years) were consecutively recruited from local Emergency Departments and Urgent Care clinics. Inclusion criteria consisted of a combination of the American Congress of Rehabilitation Medicine for mTBI (upper injury limit; Kay et al., 1993) and the Zurich Concussion in Sport Group criteria (lower injury limit; McCrory et al., 2009). Patients were examined in the sub-acute (SA;  $7.4 \pm 2.2$  days) and early chronic (EC;  $130.9 \pm 14.5$  days) stages post-injury. An age- and sex-matched group of 181 HC was recruited through word of mouth or fliers in the local community. Intervals between visits were kept consistent for both groups to account for typical neurodevelopment and repeat scanning effects.

Participants were excluded in case of the following: 1) presence of major neurologic disorders, 2) previous TBI with loss of consciousness of  $>30$  min (i.e., moderate or severe TBI), 3) developmental disorders (autism spectrum disorder or intellectual disability), 4) psychiatric disorders other than adjustment disorder, 5) contraindications for MRI (including pregnancy), 6) non-fluency in English (child or guardian) or 7) history of substance abuse/dependence. Patients with pmTBI were also excluded if they received general anesthesia during routine trauma care. Additional exclusion criteria for HC were a diagnosis of Attention Deficit Hyperactive Disorder (ADHD) or learning disability. All participants underwent urine-based drug screenings at both visits.

Follow-up data from 150 pmTBI and 164 HC were available after exclusions due to attrition, a lack of MRI data due to scanner upgrades, new psychological diagnosis, new MRI contraindications, the COVID-19 pandemic, positive drug tests or other reasons. For one patient there was only data for the EC visit, due to noise sensitivity and headaches during the SA scan. After accounting for motion outliers (see definition under Image Processing section), the final sample consisted of 200 pmTBI and 179 HC at the SA visit, and 146 pmTBI and 156 HC at EC (See Supplementary Fig. 1 for a flow chart of participant inclusion).

All study procedures were conducted according to the declaration of Helsinki. All participants provided written informed consent (i.e., parents and children aged 18 years) or assent (children less than 18 years) in accordance with the University of New Mexico School of Medicine guidelines.

### 2.2. Clinical Measures

A Common Data Elements set of neuropsychological tests and questionnaires were administered to measure cognitive functioning, post-concussive symptoms (PCS), emotional functioning, behavioral problems, quality of life, and functional outcome during both visits (see Supplementary Table 1 for a detailed description of all measures). Regarding persistent symptoms, participants were considered symptomatic if they scored  $z > 1.64$  based on ratings in HC (Mayer et al., 2020). A history of previous pmTBI was assessed using a semi-structured interview. In addition, parental distress was measured.

### 2.3. Image Acquisition

A 3T Siemens Tim Trio scanner with a 32-channel head coil was used for image acquisition (Siemens; Erlangen, Germany). See Supplementary Materials for detailed image acquisition parameters. A high-resolution MPRAGE T1-weighted scan (isotropic voxels=1 mm<sup>3</sup>) was acquired for anatomical reference. A single-shot gradient-echo-echo-planar pulse sequence (TR = 460 ms, and multiband acceleration factor = 8) was used for the acquisition of rs-fMRI (650 volumes). Participants were instructed to fixate on a centrally presented cross, to stay awake, and to think of nothing in particular. One reference functional image without multi-band imaging was acquired for registration purposes. Two spin-echo sequences with reversed-phase encoding directions (A→P; P→A) were recorded to be used for susceptibility distortion correction.

### 2.4. Image Processing

Functional images were preprocessed using the Analyses of Functional Neuroimages (AFNI; Cox, 1996) and FMRIB Software Library (FSL; Jenkinson et al., 2012) software. Data were despiked, slice timing corrected, and spatially registered (in 2D and 3D space) to the single-band reference volume. Subsequently, susceptibility distortions were corrected using FSL's *topup* function. The corrected data were orthogonalized with respect to the six motion parameters, their derivatives, the average white matter and cerebrospinal fluid signals, bandpass filtered (0.01-0.1 Hz), registered to the subject-specific T<sub>1</sub>, and non-linearly normalized to Talairach space. Subjects with mean framewise displacement (FD) higher than three times the interquartile range above the 3<sup>rd</sup> quartile of their respective group (pmTBI or HC) were excluded from further analyses (see Supplementary Fig. 1).

### 2.5. Functional Connectivity Analyses

The Automated Anatomical Labeling (AAL) atlas consisting of 90 non-cerebellar and 26 cerebellar regions (converted from MNI to Talairach space) (Schmahmann et al., 1999; Tzourio-Mazoyer et al., 2002) was used for the parcellation of functional images. Time courses per region were extracted for each subject using AFNI's *3dROIstats* function.

Dynamic FC analyses were performed using the Leading Eigenvector Dynamics Analysis (LEiDA) software (Cabral, Vidaurre, et al., 2017) implemented in MATLAB (version R2020a, The MathWorks Inc., Natick, Massachusetts, USA), and incorporated into our own customized code. For each subject, the Hilbert transform (using MATLAB's *hilbert* function) was applied to the AAL time courses to obtain the analytic signal (consisting of a real part which is the original input data, and an imaginary part that is the transform) for each region ( $n$ ) at each time point ( $t$ ). Subsequently, the phases  $\theta(n, t)$  were extracted from these analytic signals using MATLAB's *angle* function. The first and last time point were then discarded to account for edge effects (distortions at the boundaries of the analytic signal after Hilbert transform), resulting in 647 phase angles per region. To obtain a measure of dFC, the phase coherence was computed between regions  $n$  and  $p$  at each time point  $t$  using the following equation:

$$dFC(n, p, t) = \cos(\theta(n, t) - \theta(p, t))$$

This resulted in a  $[116 \times 116]$  matrix consisting of all phase coherence values (ranging from  $-1$  to  $1$ , with higher values indicating higher connectivity) per time point. To obtain the dominant pattern of FC from the phase coherence matrix at each time point ( $t$ ), the first (or leading) eigenvector ( $V_i(t)$ ) was extracted. This vector has elements that represent the BOLD phase projection of the separate AAL regions onto ( $V_i(t)$ ). These elements have either a positive or negative sign (collectively indicating the direction with the highest variance, or eigenvalue, in the FC data), delineating two communities with regions that have a similar BOLD phase relationship. As the eigenvectors  $V_i(t)$  and  $-V_i(t)$  represent the same one-dimensional space, only the relative sign is of interest. Therefore, by convention it was ensured that the majority of values within  $V_i(t)$  were negative. This is in accordance with the notion that the smallest group of regions with different BOLD phases as compared to the global mode represent meaningful brain networks (Lord et al., 2019). For each subject and visit, these analyses yielded a  $[647 \times 116]$  matrix containing all  $V_i$  for all time points. All subjects' matrices (of all visits) were concatenated in the vertical (row) dimension, resulting in a  $[440, 607 \times 116]$  matrix. Subsequently, k-means clustering (with cityblock [a.k.a. Manhattan] distance, 42 replicates, and maximum of 5000 iterations) was run using MATLAB. The Davies-Bouldin and Ray-Turi cluster validity indices were computed across a range of clusters ( $k = 3:15$ ) to find the optimal data partitioning, as reflected by local minima (Vergara et al., 2020). Two local minima were determined at  $k=5$  and  $k=10$  (see Supplementary Fig. 2). For both these local minima, a  $[440, 607 \times 1]$  vector (one for  $k=5$ , and one for  $k=10$ ) was obtained, containing the cluster membership indices indicating to which cluster each  $V_i(t)$  was assigned. A cluster will henceforth be referred to as a *dynamic brain state* or simply *brain state*. The vector representing each state  $C$  is called the centroid ( $V_c$ ) and reflects the arithmetic mean of all  $V_i$  assigned to that state.

For each subject at each visit several brain state summary measures were computed using the cluster membership indices for that subject and visit: *fractional occupancy* (defined as the number of time points assigned to a certain state divided by the total number of time points for that subject's visit; also known as the fraction of time spent, occupancy rate, or probability of occurrence), *mean dwell time* (average number of consecutive time points spent in a certain state; also known as lifetime), *number of transitions* (how many switches occur between states), and *transition probability* (the probability of switching from one state to another). These values were used for statistical analyses. The primary analyses were focused on the two local minima ( $k=5$  and  $k=10$ ), and  $p$ -values for brain state-specific measures (i.e., fractional occupancy and mean dwell time) were considered statistically significant at a Bonferroni corrected  $\alpha = 0.05/(5+10)=0.0033$ ; for transition probabilities  $\alpha$  was corrected for the total number of possible (non-self) switches across both local  $k$  minima (i.e.,  $\alpha=0.05/((5*5-5)+(10*10-10))=0.00045$ ). For number of transitions, results were considered significant at  $\alpha = 0.05/2=0.025$  (correcting for two local minima).

Although the primary analyses were focused on the two local minima, we also performed analyses for the entire range of  $k$  ( $3:15$ ), according to Fasano and colleagues (Fasano et al., 2023).

In addition to dFC, static FC was computed for each subject using Pearson's correlations between all AAL regions' entire time courses ( $116 \times 115/2 = 6670$  functional connections). Correlations were then transformed to Fisher's z-scores to increase normality. These values were used for statistical analyses. P-values were corrected for multiple comparisons using Storey's method available in MATLAB's *mafdr* function (Storey, 2002), and results were considered significant when *q*-values (corrected *p*-values) were smaller than 0.05.

## 2.6. Statistical Modeling

The Statistical Packages for Social Sciences (SPSS; IBM Corp. Released 2011. IBM SPSS Statistics for Windows, Version 20.0. Armonk, NY. USA: IBM Corp.) was used to analyze clinical and demographical data. Generalized linear models (GLM; Group effect only) or generalized estimating equations (GEE; Group and Visit effects) with Gaussian, gamma or negative binomial distributions were used, and age-at-injury (in months) and (if available) retrospective ratings were included as covariates. Results were considered significant at  $\alpha < 0.05$  with Bonferroni corrections.

To investigate differences in state summary measures, the following generalized linear mixed effects model was fit using MATLAB's *fitglme* function: [Group (pmTBI vs. HC)  $\times$  Visit (SA vs. EC)], with inclusion of mean FD as nuisance covariate and random subject intercepts, and the use of appropriate distributions (which was gamma). For static FC, similar mixed models were used (all with gaussian distributions). Results were corrected for multiple comparisons as aforementioned. In addition, Cohen's *d* effect sizes were computed.

Significant imaging findings were further examined for relationships with PCS (model 1) and injury severity measures (model 2: LOC, PTA, lesions on MRI) in the pmTBI group using linear or generalized linear models in SPSS. Additionally, with a third model the association with a history of pmTBI was examined.

## 2.7. Interpretation of Spatial State Configuration

To interpret the spatial configuration of the individual states, the 7 cortical networks defined by Yeo and colleagues were used (Yeo et al., 2011). For each of these networks, a 90-element vector was made containing the weighted contribution of the non-cerebellar AAL areas to these networks. Subsequently, Pearson's correlations were computed between these vectors and the non-cerebellar part of the state centroids (with all negative values set to zero; see Supplementary Fig. 3).

# 3. Results

## 3.1. Demographics

There were no group differences in biological sex, age, self-reported Tanner stage of development and handedness (see Supplementary Table 2). A higher rate of self-reported history of head injuries was present in the pmTBI as compared to the HC group ( $\chi^2=13.22$ ,  $p=0.004$ ; pmTBI=18.5%, HC=7.3%). Parental self-reported psychopathology ( $Wald-\chi^2=19.2$ ;  $p<0.001$ ) was also higher relative to HC. Additionally, significant group differences in premorbid reading ability ( $Wald-\chi^2=30.78$ ;  $p<0.001$ ; pmTBI<HC), and effort



( $Wald-\chi^2=9.0$ ;  $p=0.003$ ; pmTBI<HC) were found, which formed the rationale for including these variables as covariates in the analyses of neuropsychological data.

Ninety-six patients with pmTBI (48%) underwent head CT, and 7 patients in this group (7.3%) had traumatic lesions.

### 3.2. Clinical analyses

Table 1 lists all data for self-reported clinical outcomes and neuropsychological measures. A significant Group $\times$ Visit interaction ( $Wald-\chi^2=17.67$ ;  $p<0.001$ ) was observed for PCS severity ( $p < Bonferroni$  corrected  $\alpha=0.0167$ ). Follow-up GLM analyses showed that PCS severity was higher in pmTBI relative to HC at both the SA ( $p<0.001$ ; 34% of pmTBI classified as symptomatic based on (Mayer et al., 2020)) and EC visits ( $p<0.001$ ; 16% of pmTBI symptomatic). No significant effects were found for conflict behavior and the primary quality of life outcome measures.

For the secondary clinical measures of sleep, pain and headaches a significant Group $\times$ Visit interaction was present (all  $p$ 's  $< Bonferroni$  adjusted  $\alpha=0.007$ ). Follow-up tests showed more sleep problems in the pmTBI group relative to HC at SA ( $p<0.001$ ) and less pronounced at the EC ( $p<0.05$ ) visit, and increased pain and headaches at the SA ( $p$ 's $<0.001$ ) but not EC visit. For depression, anxiety, functional outcome (GOS-E), and a behavioral screening measure (Strengths and Difficulties Questionnaire) a significant main Group effect was found (all  $p$ 's $<0.001$ ; pmTBI>HC, indicating worse symptoms or outcome).

Regarding the primary measures of cognitive functioning, a significant Group $\times$ Visit interaction was observed for processing speed ( $Wald-\chi^2=5.32$ ,  $p=0.021 < adjusted \alpha=0.025$ ) and attention ( $Wald-\chi^2=6.17$ ,  $p=0.013 < adjusted \alpha=0.025$ ), with lower values in pmTBI relative to HC during the SA, but not EC visit (indicating worse performance). A significant main effect of Group (pmTBI<HC) was found for the secondary measures of executive functioning ( $p=0.006$ ) and delayed recall ( $p<0.001$ ). Groups did not differ on working memory performance.

### 3.3. Dynamic FC results

Effects of pmTBI on dynamic FC were found when the data was partitioned into 10 states (the second local minimum; see Fig. 1 for the spatial characteristics of these states). Cortical networks defined by Yeo and colleagues (Yeo et al., 2011) were used to interpret the spatial configuration of the positive communities (see Supplementary Fig. 3). State 1 and 5 were labeled as DMN states, and state 9 as a DMN/limbic state. State 2 (with inclusion of thalamus/caudate areas) and 8 (with inclusion of paracentral areas) were labeled as cerebellar states, and state 3 as a sensorimotor state. State 4 was labeled as a visual/dorsal attention state, and state 7 as a sensorimotor/frontoparietal/ventral attention state. State 6 represents a frontoparietal/dorsal attention state. Finally, state 10 was labeled as visual state (with inclusion of cerebellar areas).

Generalized mixed effects models ([Group (pmTBI vs. HC)  $\times$  Visit (SA vs. EC)]) revealed a significant main effect of Group for fractional occupancy for state 9 ( $t=-4.19$ ,  $p < 0.0001$ ).



< Bonferroni corrected  $\alpha = 0.0033$ ; pmTBI<HC; Fig. 2). This DMN/limbic state was characterized by key nodes within the basal and midline frontal regions, temporal areas, as well as in the amygdala, hippocampus and paracentral sensory areas. It can be noted that in contrast to the other DMN states (1 and 5), state 9 has limited nodes within the parietal midline (precuneus or posterior cingulate). Cohen's  $d$  effect sizes for SA ( $d_{SA}$ ) and EC ( $d_{EC}$ ) were  $-0.32$  and  $-0.42$ , respectively. The Group effect for this specific state was present across all  $k$  examined from  $k=10$  up to  $k=15$ , which shows the robustness of this finding (Supplementary Fig. 4). A trend for significantly decreased mean dwell time was found for this same state in pmTBI versus HC ( $t=-1.72$ ,  $p=0.086$ ). For number of state transitions, no significant effects were observed. None of the effects regarding transition probabilities survived the Bonferroni corrected threshold ( $\alpha = 0.05/110 = 0.00045$ ).

When partitioning the brain into 5 states (the first local minimum; Supplementary Fig. 5), no significant Group or Group $\times$ Visit effects were observed for the state summary measures.

Within the pmTBI group, no significant associations were found between fractional occupancy of state 9 and either PCS or injury severity measures (LOC, PTA, lesions on MRI). Furthermore, there was no relationship with a history of previous pmTBI.

### 3.4. Static FC results

Linear mixed models ([Group (pmTBI vs. HC)  $\times$  Visit (SA vs. EC)]) revealed significant main effects of Group for static FC between the right supplementary motor area (rSMA) and left cerebellar area 10 ( $t=-4.85$ ,  $q=0.0078$ ; pmTBI < HC,  $d_{SA}=-0.28$  and  $d_{EC}=-0.51$ ), between left and right precuneus ( $t=4.26$ ,  $q=0.0399$ ; pmTBI>HC,  $d_{SA}=0.35$  and  $d_{EC}=0.31$ ), and between left and right thalamus ( $t=4.39$ ,  $q=0.0335$ ; pmTBI>HC,  $d_{SA}=0.36$  and  $d_{EC}=0.41$ ) following stringent corrections for false positives (Fig. 3B and C).

Within the pmTBI group, no significant associations were found between static FC for these 3 connections and clinical measures, or previous mTBI.

## 4. Discussion

To our knowledge, this is the first rs-fMRI study investigating dFC after pmTBI. In a large longitudinal study sample, we showed that patients with pmTBI spend less time in a frontotemporal DMN/limbic dynamic brain state relative to HC, lasting up to four months post-injury. Furthermore, we found altered static interhemispheric connectivity between left and right precuneus and thalamus, and between right supplementary motor area and contralateral cerebellum. Although patients reported significantly higher levels of PCS at both visits as compared with HC, this was not related to pmTBI-induced changes in static or dynamic FC measures. Our findings also did not relate to injury severity measures, including PTA and LOC.

Given the complexity of the human brain it is not surprising that the effects of mTBI are still far from understood, let alone in children and adolescents whose brains are still in development. Although fMRI has provided a means to gather insight into pmTBI (Mayer et al., 2018), studies so far have not examined the more dynamic aspects of brain connectivity.

In the current study, we revealed altered brain dynamics after pmTBI, as reflected by less time spent in a frontotemporal DMN/limbic state. Areas that were most strongly incorporated in this affected brain state were the orbito- and mediofrontal cortex as well as the temporal poles. These areas are especially vulnerable to TBI due to the irregular morphology of the skull base in the anterior and temporal cranial fossa, as well as the maximum strain and strain rates on the frontal midline (McAllister, 2011; McAllister et al., 2012). Therefore, it is plausible that damage to these regions translates into difficulty engaging in a frontotemporal brain state, resulting in lower fractional occupancy in this state following pmTBI. In contrast to our a priori hypotheses, this effect was present up to four months post-injury. This finding warrants more research to determine the long-term effects on neurodevelopment. An additional potential avenue for further research would be to employ dFC analyses to task-based fMRI data obtained from children and adolescents with mTBI, potentially unveiling more subtle alterations in brain dynamics associated with more complex cognitive functioning (e.g., working memory, cognitive control (van der Horn et al., 2023)).

The DMN, with core areas located within the prefrontal and parietal midline, is associated with internal mentation and is deactivated during externally goal directed attention (Raichle, 2015). The DMN has been further subdivided into core, medial temporal, and dorsomedial subsystems (Andrews-Hanna et al., 2014). However, the configuration of a dynamic brain state is not confined to one of the canonical brain networks or their subsystems, but may incorporate areas from different networks (Alonso Martinez et al., 2020; Figueroa et al., 2019; Lord et al., 2019). In the current study, three DMN states were observed, but only a single state was affected by pmTBI. The orbitofrontal and temporal DMN/limbic state that was specifically affected by pmTBI included the amygdalae, with a previous network parcellation (Yeo et al., 2011) also labeling the regions as a limbic state. Although regions within this specific state have been associated in the past with emotion regulation (Ochsner & Gross, 2005), current findings were not associated with persistent PCS. It seems likely that other (subtle and more complex-weighted) factors determine whether or not symptoms develop and/or persist (Mayer et al., 2017; Polinder et al., 2018; Zemek et al., 2016), which cannot be accurately measured using rs-fMRI, or with the (psychometric) screening instruments that were used in the current study. Possible explanations for not finding relationships with injury severity measures are that perturbed functional connections underlying transient phenomena such as LOC and PTA might have been recovered at a week post-injury in mild TBI (in contrast to moderate-severe TBI; De Simoni et al., 2016), or that more focal effects within the ascending reticular activating system (Jang et al., 2019) could not be measured using rs-fMRI.

Our static FC analyses revealed altered interhemispheric connectivity between the left and right precuneus, thalamus, and between the right supplementary motor area and left cerebellum. Our findings are supported by previous studies in adult and pediatric mTBI that have shown altered functional and structural connectivity of these regions (Aoki & Inokuchi, 2016; Murdaugh et al., 2018; Sours et al., 2015; Tang et al., 2011; Vergara et al., 2017; Vergara et al., 2018). Regarding the direction of FC changes, increased functional connectivity between left and right precuneus and thalamus might be a compensatory phenomenon (Manning et al., 2017; Murdaugh et al., 2018), whereas decreased connectivity

between cortex and cerebellum might reflect cortico-cerebellar decoupling (Hattori et al., 2009). The fact that midline connections were affected is consistent with neuromechanical models of TBI and suggests the accumulation of shear stresses in these regions (McAllister, 2011; McAllister et al., 2012). The corpus callosum is one of the main structural pathways via which the precuneus (Tanglay et al., 2022) and thalamus (Johnston et al., 2008) communicate with their contralateral homologues, and this white matter structure is particularly susceptible to traumatic injury (Aoki & Inokuchi, 2016; Wang et al., 2021). Similarly, frontocerebellar connections are vulnerable as they run close to the midline, via or adjacent to the thalamus (Palesi et al., 2017), and the corticospinal tract (Murdaugh et al., 2018) and cerebellar peduncles have found to be affected by mTBI (Mallott et al., 2019).

Because of its simplicity, the k-means algorithm is a widely adopted data clustering method (Ikotun et al., 2023), but one has to select a specific number of clusters ( $k$ ). Interestingly, in the current study effects of pmTBI on dFC were only revealed when the data was partitioned into a higher number of brain states, which underlines the importance of choosing the right partitioning structure for uncovering injury-related effects. One could draw a parallel here with the model order selection in independent component analysis (ICA), which is a data-driven technique used for the parcellation of fMRI data (Abou Elseoud et al., 2011). The number of clusters in the current study was determined using local minima of cluster validity indices (Vergara et al., 2020). A previous study has advocated to examine the whole range of  $k$ , instead of only optimal partitions based on cluster validity indices, with the aim of finding the most discriminatory data partitioning for a particular research question (Fasano et al., 2023). However, this method might be prone to type I errors, and therefore we only applied this as a secondary analysis. Nevertheless, this analysis revealed that our dFC findings at the  $k=10$  clustering optimum remained robust throughout even higher clustering partitions.

Our study has several strengths and limitations that are worth mentioning. Foremost, we prospectively sampled one of the largest cohorts of pmTBI and matched HC to date, and acquired longitudinal rs-fMRI data that allowed us to investigate recovery patterns. The sample was demographically as well as clinically well-characterized, and we had excellent (pmTBI: 81%; HC: 94%) retention rates. A possible limitation is the chosen brain parcellation method (i.e., AAL1 atlas). Although used frequently with the LEiDA method (Cabral, Vidaurre, et al., 2017; Fasano et al., 2023; Figueroa et al., 2019; Hancock et al., 2022; Lord et al., 2019), it might not be fine-grained enough to capture every aspect of dFC after pmTBI. Future studies might benefit from comparing results obtained with multiple atlases. Second, our rs-fMRI scanning duration was relatively short with the aim of reducing potential motion artifacts in children (White et al., 2014), and this might have reduced the reliability of connectivity measures (Birn et al., 2013). However, part of this was mitigated by using a multi-band acquisition sequence (repetition time 460 ms), resulting in an excellent temporal resolution, and by applying an analysis method that captures instantaneous FC without the need for a sliding window. Lastly, although the brain states aligned well with previously published canonical functional brain networks (Yeo et al., 2011), it goes without saying that the brain is not limited to 5 or 10 dynamic states. For understanding the effects of pmTBI more novel methods may be needed.

In conclusion, the current study suggests changes in dFC after pmTBI, more specifically affecting the engagement in a frontotemporal DMN/limbic state. The clinical implication of this finding, other than it being caused by trauma, is uncertain. Future studies are needed to elucidate how brain state dynamics relate to patients' cognitive and emotional wellbeing, and to their neurodevelopmental trajectory. The relationship with subjective wellbeing is likely to be complex, and the best clinical measures to capture the heterogeneous nature of PCS have yet to be determined. Our study has also demonstrated that static FC of midline connections is affected by pmTBI, which fits with known neurobiological models of TBI.

## Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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## Data availability

The data that support the findings of this study will be openly available in FITBIR at [fitbir.nih.gov](http://fitbir.nih.gov) upon the conclusion of the study, reference number FITBIR-STUDY0000339.

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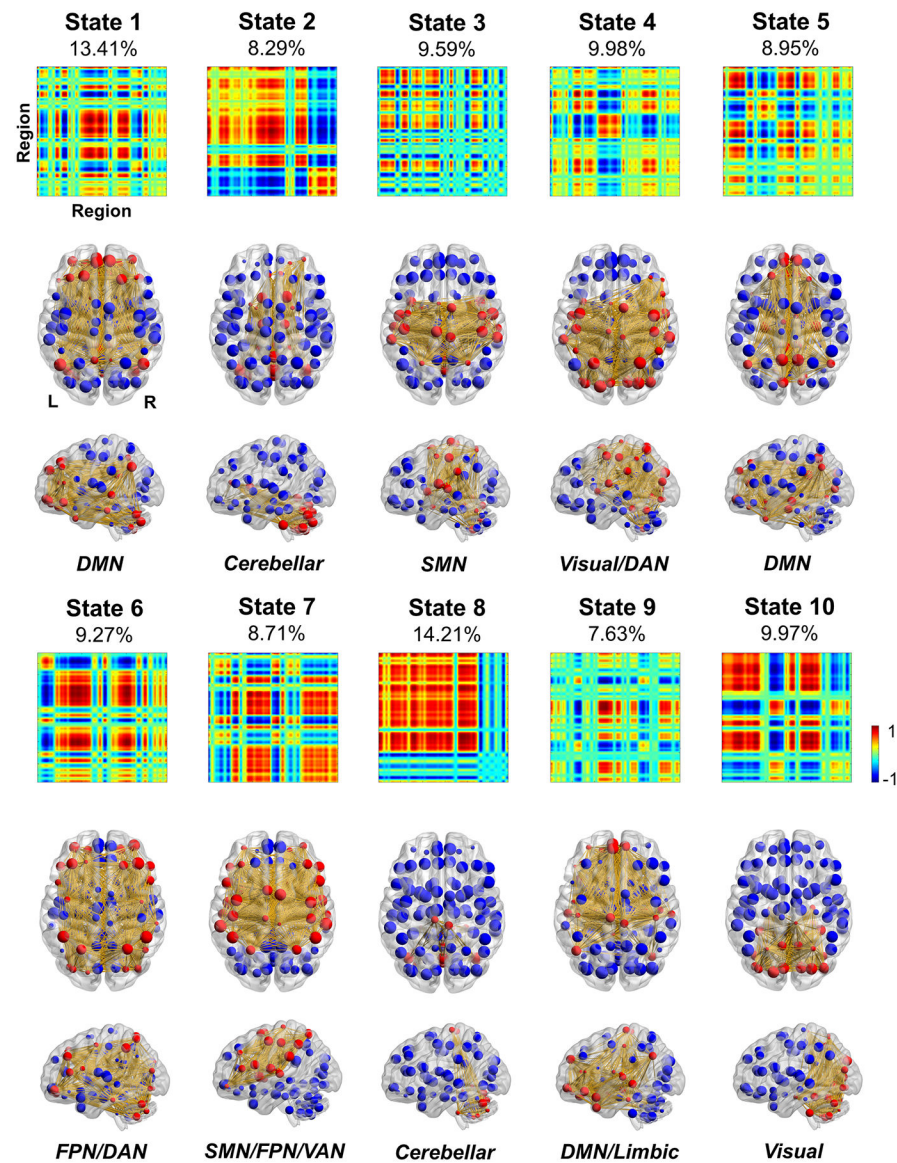
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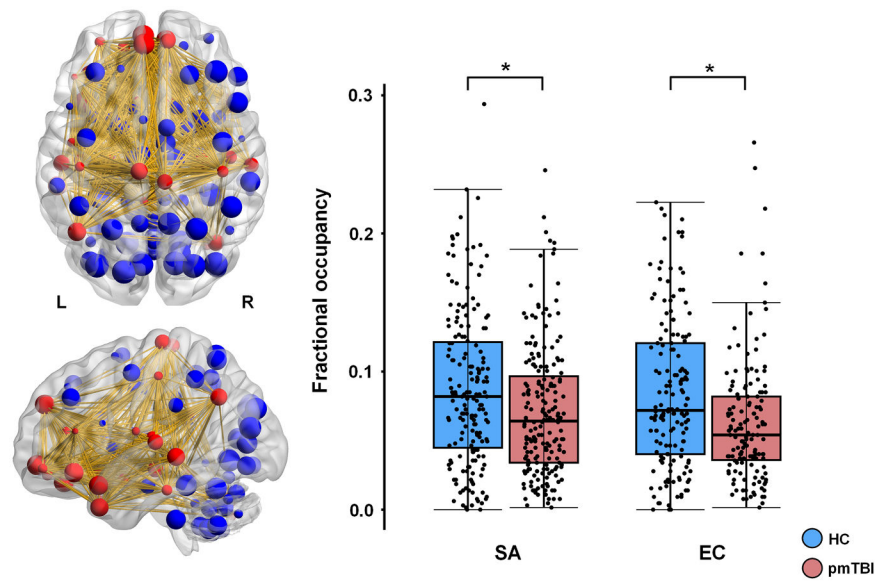


**Fig. 1.**

Individual brain state configurations for the  $k=10$  partition. Matrix plots show the outer products ( $V_c V_c^T$ ) of the state centroids  $V_c$ , with red matrix elements indicating pairs of nodes (i.e., AAL regions on axes in sequential order) that have similar sign within  $V_c$  (nodes that belong to the same community), and blue elements indicating nodes with opposite sign (i.e., nodes that belong to different communities). Because the leading eigenvector at a specific point in time ( $V_1(t)$ ) denotes a direction in the functional connectivity data, it can have positive and negative values, and therefore there are two possible communities for each time point.  $V_c$  represents the arithmetic mean (across all participants) of all timepoints ( $V_1(t)$ ) assigned to a state by k-means clustering, and therefore consists of two (average) communities. Above the matrices the percentage of total time (across all participants) that was assigned to these states is depicted. Rendered brain images show nodes (i.e., AAL regions) that are weighted by their absolute value in  $V_c$  and colored based on their sign

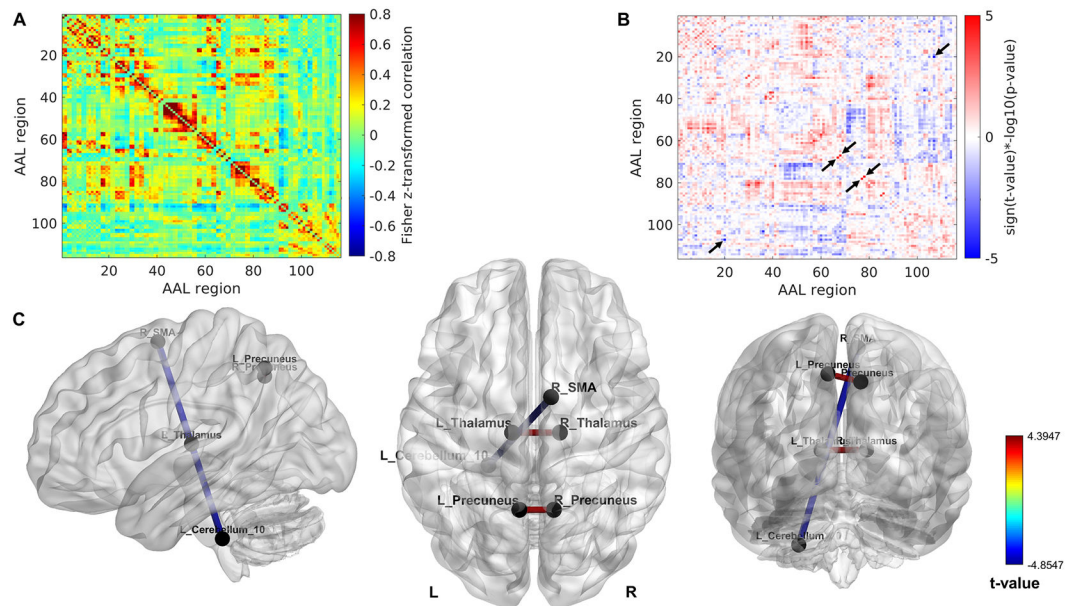
in  $V_c$  (red = community with positive values in  $V_c$ , and blue = negative community; by convention the smallest community is positive and forms a network [hence the edges] with different BOLD phase relationship as compared to the global mode). Figures were made using custom MATLAB scripts and BrainNet Viewer (Xia et al., 2013).

Abbreviations: DAN = dorsal attention network; DMN = default mode network; FPN = frontoparietal network; SMN = sensorimotor network; VAN = ventral attention network.



**Fig. 2.**

Fractional occupancy for state 9. Patients had significantly lower values (marginal mean = 0.067, SD = 0.055) across the subacute (SA) and early chronic (EC) visits as compared to HC (marginal mean = 0.086, SD = 0.055). Asterisks indicate  $P < 0.05$  after correction for multiple comparisons. Figures were made using custom MATLAB scripts and using BrainNet Viewer (Xia et al., 2013); boxplots were made using the *ggplot2* package in R (R Core Team 2023, Vienna, Austria).

**Fig. 3.**

Static FC results. (A) Mean static FC matrix for the total group of participants. (B) Differences in static FC between pmTBI and HC as expressed by the negative  $\log_{10}$  of p-values multiplied by the sign of t-values. The direction of the group difference is denoted by the color of the element in the matrix (red = pmTBI > HC; blue = pmTBI < HC), and the darker the color the more significant the group difference is. Arrows show the significant group differences at  $q < 0.05$  ( $q = \text{adjusted } p\text{-value}$ ); (C) Connections that were significantly different between groups (corresponding with arrows in B), with blue rods indicating decreased FC, and red rods increased in pmTBI relative to HC. Thus, the colors of the rods correspond with the colors of the elements of the matrix in panel B. Figures were made using custom MATLAB scripts and BrainNet Viewer (Xia et al., 2013).

**Table 1**

Clinical and neuropsychological data for both groups at both visits.

| Metric                                   | Outcome | SA pmTBI<br>(N=200) | SA HC<br>(N=179) | EC pmTBI<br>(N=146) | EC HC<br>(N=156) |
|------------------------------------------|---------|---------------------|------------------|---------------------|------------------|
| <b>Symptom Measures</b>                  |         |                     |                  |                     |                  |
| PCSI (% Max) <sup>b</sup>                | P       | 13.5(4.8-32.5)      | 2.9(0-9.2)       | 4(0-15.9)           | 3.2(0.8-7.5)     |
| PROMIS                                   | S       | 19(13-24)           | 13(11-16)        | 18(12-23)           | 14(11-18)        |
| Sleep <sup>b</sup>                       |         |                     |                  |                     |                  |
| PROMIS                                   | S       | 3(0-8)              | 1(0-4.5)         | 2(0-6)              | 1(0-4.5)         |
| Anxiety <sup>a</sup>                     |         |                     |                  |                     |                  |
| PROMIS                                   | S       | 2(0-8)              | 0(0-3.5)         | 1(0-5)              | 1(0-3)           |
| Depression <sup>a</sup>                  |         |                     |                  |                     |                  |
| Pain Scale <sup>b</sup>                  | S       | 3(1-5)              | 0(0-1)           | 0(0-2)              | 0(0-1)           |
| HIT-6 <sup>b</sup>                       | S       | 52(44-59)           | 40(36-47)        | 44(40-55)           | 42(38-47)        |
| <b>Behavioral &amp; Outcome Measures</b> |         |                     |                  |                     |                  |
| CBQ                                      | P       | 1(0-2)              | 1(0-2)           | 1(0-3)              | 1(0-2)           |
| PedsQL                                   | P       | N/A                 | N/A              | 83.4±14.1           | 87.7±10.2        |
| SDQ <sup>a</sup>                         | S       | N/A                 | N/A              | 7.7±5.5             | 5.3±4.4          |
| GOS-E <sup>a</sup>                       | S       | 1.5(1-4)            | 1(1-1)           | 1(1-2)              | 1(1-1)           |
| <b>Cognitive Measures</b>                |         |                     |                  |                     |                  |
| TOMMe10 <sup>a</sup>                     | S       | 10(9-10)            | 10(10-10)        | 10(10-10)           | 10(10-10)        |
| WRAT4 <sup>a</sup>                       | S       | 49(43.3-54.7)       | 54(47.7-62.4)    | 49.3(43.3-56)       | 55.3(49.3-62.7)  |
| PS <sup>b</sup>                          | P       | 47.1±7.5            | 50.4±8.1         | 50.1±8.4            | 52.4±8.8         |
| AT <sup>b</sup>                          | P       | 47.8(42.2-53.3)     | 52.2(46.7-55.6)  | 50(44.5-55)         | 52.2(47.8-56.7)  |
| WM                                       | S       | 46.7(43.3-50)       | 46.7(43.3-53.3)  | 46.7(43.3-56.7)     | 50(43.3-56.7)    |
| EF <sup>a</sup>                          | S       | 47.7±7.1            | 50.3±6.7         | 50.1±6.8            | 52.9±6.4         |
| HVLT Delay <sup>a</sup>                  | S       | 8(6-10)             | 9(7-10)          | 7(6-9)              | 9(7-10)          |

Notes: Outcomes are classified as primary (P) or secondary (S). Acronyms: SA=sub-acute; EC=early chronic; HC=healthy control; N/A = not applicable; pmTBI=pediatric mild traumatic brain injury; PCSI= Post-Concussion Symptom Inventory (presented as percent of maximum score to account for age-related scale differences); PROMIS= Patient Reported Outcomes Measurement Information System; HIT-6=headache impact test; CBQ= Conflict Behavior Questionnaire; PedsQL= Pediatric Quality of Life Inventory; SDQ= Strengths and Difficulties Questionnaire; GOS-E= Glasgow Outcome Scale Extended; TOMMe10= Test of Memory Malingering – 10-item short version; WRAT4= Wide Range Achievement Test 4; PS=processing speed; AT=attention; WM=working memory; EF=executive function; HVLT Delay = Delayed recall on Hopkins Verbal Learning Task (measure of long-term memory). Data are either formatted at mean ± standard deviation or median (interquartile range).

<sup>a</sup>Group main effect.

<sup>b</sup>Visit interaction.